

Responds to the reviewers' comments on TMECH Submission 02-2026-23812

Dear Prof. Paolo Robuffo Giordano,

Our paper, TMECH Submission 02-2026-23812, "Vision-Based Control of Flexible Joint Manipulators Using Motor Position Measurements", has been revised according to the Editor's and each reviewer's comments. The itemized replies to the comments and the changes made in the revised paper are attached. Many thanks for your careful review and valuable processing of our paper.

Yours sincerely,

Fan Xu, Kai Li, Lijun Han, Hesheng Wang

Our replies to the Editor's and each reviewer's comments are appended as below.

The replies to the Editor's comments

Dear Senior Editor,

Thank you very much for reviewing carefully and making valuable criticisms and helpful suggestions on our paper.

According to your comments:

Both TE and reviewer raise questions and concerns which need to be addressed. As summarized in TE's comments, this paper needs major revision.

Our reply and revision work: We sincerely thank your thorough evaluation of our manuscript and for the many constructive and insightful comments. Following the reviewers' and Editorial Board's recommendations, we have carefully revised the manuscript. We hope that the revised manuscript satisfactorily address all concerns and meet the high standards of the journal. We would be grateful for the opportunity to have our revised manuscript reconsidered for publication.

Yours sincerely,

Fan Xu, Kai Li, Lijun Han, Hesheng Wang

Dear Technical Editor,

Thank you very much for reviewing carefully and making valuable criticisms and helpful suggestions on our paper.

According to your comments:

The reviewer has raised several significant concerns regarding the revised manuscript, including issues related to the adequacy of experimental validation and instances of overstatement (e.g., claims on practicality). These concerns should be carefully addressed in a further revision. The authors are requested to submit a revised manuscript accompanied by a detailed response letter that clearly and systematically addresses each of the reviewer's comments.

Our reply and revision work: We sincerely thank you for the thorough evaluation of our manuscript and for the many constructive and insightful comments. Following the Editorial Board's and reviewers' suggestions, we have carefully revised the manuscript to substantially improve its clarity of presentation, and experimental thoroughness. We hope that the revised manuscript address all concerns and meet the high standards of the journal. We would be grateful for the opportunity to have our revised manuscript reconsidered for publication.

Yours sincerely,

Fan Xu, Kai Li, Lijun Han, Hesheng Wang

The replies to the Reviewer's comments

Dear Reviewer 1,

Comment 1: My previous concerns have been addressed in this version. No more questions from my side.

Our reply: We sincerely thank you for your rigorous review, valuable professional suggestions, and recognition of our revised manuscript. Your feedback has greatly helped us improve the overall quality of this work, and we are truly grateful for your time and effort dedicated to the review process.

Dear Reviewer 2,

Comment 1: The experimental validation of the adaptive visual servoing framework appears insufficient. It is unclear whether the selected feature point triplet ([267, 96], [302, 166], and [176, 146]) is adequate to support the generality of the proposed method.

Our reply: We sincerely thank the reviewer for raising this important point regarding the generality and adequacy of our experimental validation. We appreciate the opportunity to clarify the theoretical rationale behind our feature selection. From a theoretical perspective, the selection of the feature point triplet is deeply grounded in visual servoing kinematics. In image-based visual servoing (IBVS), the interaction matrix (or image Jacobian) maps the spatial velocity of the end-effector to the velocity of the features in the image plane. Because each image feature point provides two independent constraints (the u and v coordinates), tracking three feature points yields 6 independent geometric equations. As long as these three points are non-collinear in 3D space, the resulting stacked $6 \times n$ interaction matrix maintains full rank, which is theoretically sufficient to uniquely constrain the spatial pose and motion of the rigid body locally. In our experimental setup, the three selected feature points were placed in a non-collinear configuration on the physical end-effector. Consequently, their initial image projections, [267, 96], [302, 166], and [176, 146], form a non-collinear triangle. As established in our Lyapunov stability analysis (Theorem 2), when the feature points are fixed to a rigid body on the end-effector, the image position error is guaranteed to asymptotically converge to zero ($\lim_{t \rightarrow \infty} \Delta y_i = 0$). Therefore, the successful regulation demonstrated with this specific triplet is not an isolated case, but rather a direct reflection of the theoretical guarantees provided by the proposed control framework.

We fully agree with the reviewer that this underlying rationale should be made transparent to the readers to explicitly support the generality of our method. To address this, we have revised Section II.A of the manuscript to include a remark (**Remark 1**) on feature selection and have added relevant visual servoing references (**Ref.[xxx]**) to substantiate this standard practice.

Comment 2: In Fig. 9 (b), the initial positions of the three selected features appear relatively close to the image centre in the u - v plane. Under the assumption of unknown intrinsics, the performance validation experiments in Section V.A should be repeated with diverse camera poses and varying feature point configurations. Furthermore, the results of such repeated experiments should be explicitly reported within the same section.

Our reply:

Comment 3: On page 2 (Introduction), the manuscript states: “Moreover, conventional VS schemes typically rely on accurate offline camera calibration...”. This claim appears overstated. In conventional visual servoing studies, monocular cameras, standard toolchains, and widely available libraries are frequently employed. Intrinsic calibration can be readily performed using checkerboard patterns, publicly available toolkits, and minimal data acquisition. For instance, OpenCV and MATLAB provide well-established functions and accessible implementations for camera calibration. Therefore, the necessity of replacing offline calibration with the proposed approach should be more rigorously justified and supported by appropriate references.

Our reply: We sincerely thank the reviewer for this insightful comment. We completely agree that in conventional laboratory settings, offline camera calibration is a mature, accessible process made straightforward by standard li-

baries such as OpenCV and MATLAB. We apologize if our original phrasing inadvertently overstated the difficulty of this standard procedure. Following your constructive feedback, we have revised the manuscript to properly contextualize this aspect of our work. In the revised manuscript, we have modified this paragraph to soften the claim regarding conventional VS schemes.

First, we wish to clarify that replacing offline intrinsic calibration is not the primary contribution of this manuscript. The core innovation is the design of a unified singular-perturbation (SP) control architecture that concurrently achieves task-space visual servoing (in the slow subsystem) and elastic-joint vibration suppression (in the fast subsystem), utilizing exclusively motor-side position measurements and image feedback, as summarized at the end of Section I. Within this framework, the assumption of uncalibrated intrinsics is introduced primarily to demonstrate the capacity of the proposed controller under *relaxed constraints*, rather than acting as the sole contribution. While standard offline calibration is highly accessible, it requires a dedicated phase involving specific physical targets (e.g., checkerboard patterns) and repetitive multi-pose image acquisition. By integrating online intrinsic parameter estimation, the procedure becomes target-free and somehow can simplify the deployment workflow in unstructured environments or on platforms where executing a checkerboard calibration protocol prior to operation is inconvenient.

Following the reviewer’s constructive feedback, we have modified the paragraph in the Introduction to soften our claims regarding conventional VS schemes. We have reframed the inclusion of online intrinsic estimation as a means to relax operational constraints, subordinating it to our primary contribution of state-estimation-based vibration suppression. Furthermore, we have incorporated appropriate references [19, 20] discussing target-free uncalibrated VS deployment to justify this design choice.

Comment 4: Although intrinsic parameters are estimated within the study, their accuracy is not sufficiently validated. A direct comparison with parameters obtained via standard offline calibration should be included, and discrepancies should be quantitatively analysed.

Our reply:

Comment 5: The manuscript does not provide sufficient details regarding the experimental setup. The camera model, resolution, and field of view (FoV) are not specified. Additionally, it remains unclear whether the selected feature point configuration is optimal for the given camera frame. The authors are encouraged to discuss whether variations in feature spacing were considered and how the setup was optimised. The practical aspects of the system should be described in greater detail.

Our reply: We apologize for omitting the specific hardware parameters in the original manuscript. In the revised Section V, we have explicitly added the missing specifications. The flexible link is driven by a DC servo motor manufactured by Haitai Electromechanical Equipment Co., Ltd. Visual feedback is provided by a camera from Shenzhen Zhongwei Aoke Technology Co., Ltd., operating at a resolution of [Insert Resolution, e.g., 640×480] pixels with a Field of View (FoV) of approximately [Insert FoV, e.g., 70°]. Both the image processing algorithm and the proposed controller are implemented on a standard PC.

Regarding the optimality of the feature point configuration and camera placement, we wish to clarify that in standard Image-Based Visual Servoing (IBVS), finding an optimal feature spacing is not necessary to execute a manipulation task or guarantee convergence. As established in classical visual servoing literature [23], the fundamental

geometric requirement to realize 6-DOF control of an end-effector is simply that the selected feature points are non-collinear in 3D space. Therefore, our configuration was chosen to satisfy this fundamental mathematical condition rather than to pursue a specific numerical optimum. Similarly, the setup procedures and feature arrangement were not optimized, but rather configured pragmatically according to the manipulation task, which is standard practice in applied IBVS. In general, camera placement and feature spacing in IBVS are determined by two primary practical constraints. **1.** Ensuring that all feature points remain within the camera’s FoV throughout the entire operational trajectory. In practical VS operations, maintaining visibility can be guided by empirical experience. A common approach is to configure the camera and feature placements such that both the initial and desired feature coordinates are kept away from the image boundaries. **2.** Ensuring that the features maintain sufficient spatial spread (i.e., definitive non-collinearity) relative to the camera distance. This prevents the interaction matrix from becoming ill-conditioned, which could otherwise amplify system noise and degrade control performance.

More details related to VS can be found in [22]. To ensure the experiments clear to the readers, we have expanded the first paragraph of Section V.A. We have included the exact camera hardware specifications and added a brief discussion explaining that the feature setup and camera placement, in accordance with standard IBVS deployment practices.

Comment 6: The abstract claims practical applicability, and this claim is reiterated throughout the manuscript. However, no concrete example from an industrial domain or real-world application is provided. While the theoretical contribution is acknowledged, the practical relevance remains insufficiently substantiated.

Our reply: We sincerely thank the reviewer for this constructive comment on the practical value of our work. We have revised the manuscript to substantiate the practical applicability of the proposed method, also stated as follows. **We have revised the manuscript to bridge the gap between our theoretical contributions and their real-world engineering potential.**

The core practical problem our method resolves is the reliance on link-side encoders and velocity sensors. By achieving adaptive VS and vibration suppression using only motor-shaft position, our approach aligns with the hardware constraints of several industrial and real-world scenarios. For instance, flexible-joint surgical robots frequently operate in narrow, unstructured in vivo environments where embedding multiple link-side sensors is physically impractical due to strict dimensional limitations and rigorous sterilization constraints, and where dynamic surgical scenes are resistant to offline pre-calibration. Similarly, in space, orbital, and extreme environments, such as deep space exploration, nuclear decommissioning, or underwater operations, manipulators are subjected to payload and wiring limitations, elevated risks of sensor failure, and conditions that render manual offline camera calibration unfeasible. In these hostile domains, the ability of the proposed framework to effectively damp elastic vibrations with minimal sensor reliance demonstrates practical value. Furthermore, by eliminating the necessity for expensive dual-encoder configurations and rigorous factory visual calibration, this method facilitates the development of accessible, low-cost collaborative arms, reducing overall hardware and deployment costs while enhancing the rapid-deployment flexibility of commercial systems.

We acknowledge the reviewer’s underlying point that this framework has not yet been deployed on an actual commercial production line. As is typical for control theory research at this stage, the primary focus of the current manuscript is establishing the theoretical and algorithmic foundations necessary to handle complex coupled dynamics under restricted sensing conditions. Validating the method in a controlled laboratory setup is a prerequisite. To

translate this theoretical potential into direct industrial impact, our future research will focus on the actual deployment of this framework in production and industrial environments, targeting the robotics scenarios mentioned above. To ensure this practical relevance is clearly communicated to the readers, we have expanded the discussion to outline our future plans for industrial deployment. We hope these additions can address your concerns and better substantiate the practical relevance of our work.

The added relevant discussion is attached as follows.

“The proposed control framework exhibits direct application potential in the scenarios where hardware constraints and unstructured deployment scenarios preclude the use of redundant link-side sensors and reducing the deployment cost. While the primary focus of this study establishes the theoretical foundations for sensing-constrained flexible-joint manipulators, translating this potential into direct industrial impact remains a critical next step. Future research will prioritize the practical deployment of this framework in real-world scenarios, specifically targeting compact surgical robots, extreme-environment manipulators, and low-cost collaborative arms. Our future research will focus on the practical engineering deployment and performance optimization of this framework in real industrial production lines, to further validate and enhance its practical applicability in complex real-world operation scenarios.”

At last, We sincerely appreciate your valuable comments and careful review of our manuscript. We have responded to your suggestions point by point and completed the corresponding revisions. We hope the revised manuscript meets your requirements.

Dear Reviewer 3,

Comment 1: The authors have thoroughly revised the manuscript based on the revision suggestions, and I have no further comments.

Our reply: We sincerely thank you for your rigorous review, valuable professional suggestions, and recognition of our revised manuscript. Your feedback has greatly helped us improve the overall quality of this work, and we are truly grateful for your time and effort dedicated to the review process.